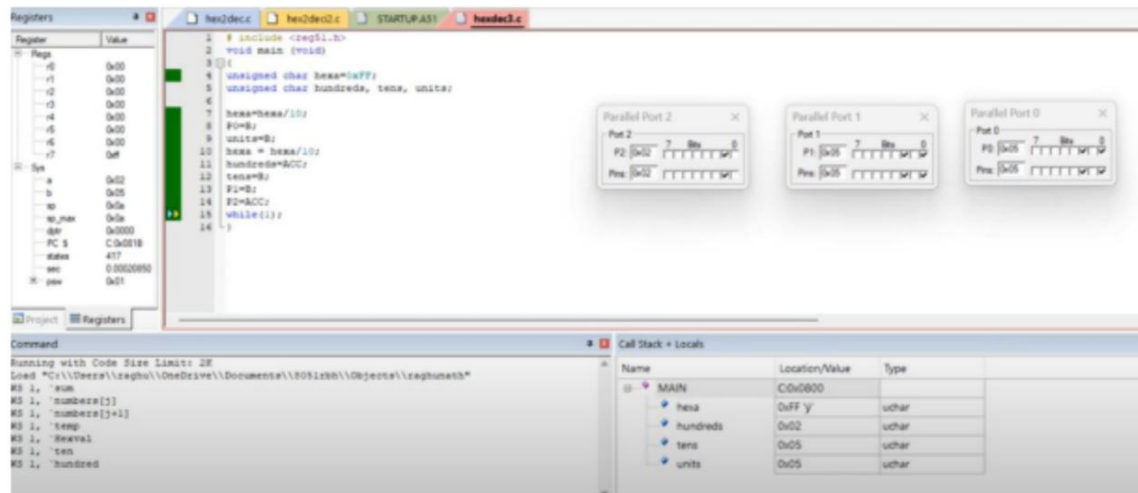
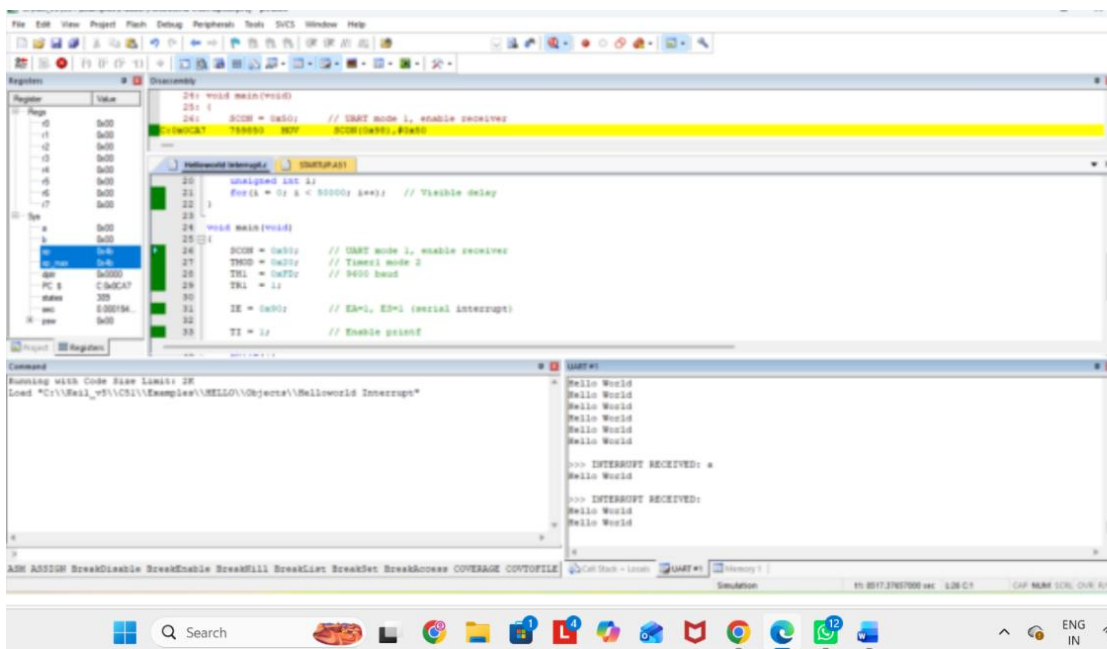


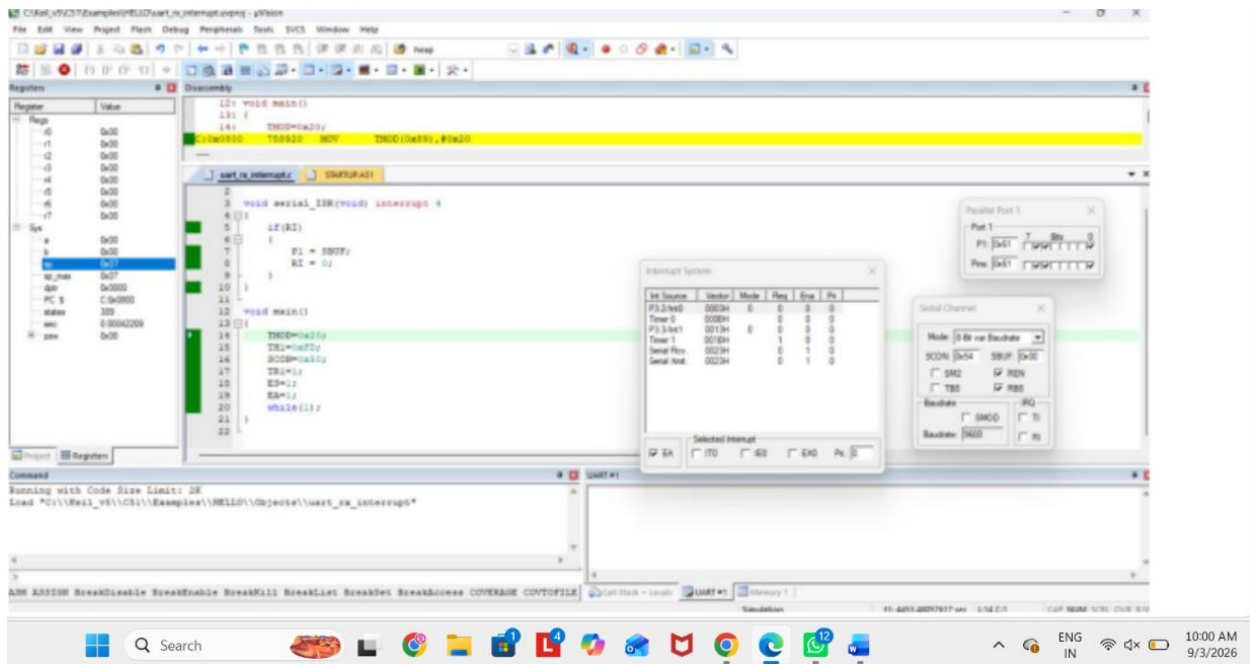
EXPERIMENT :15



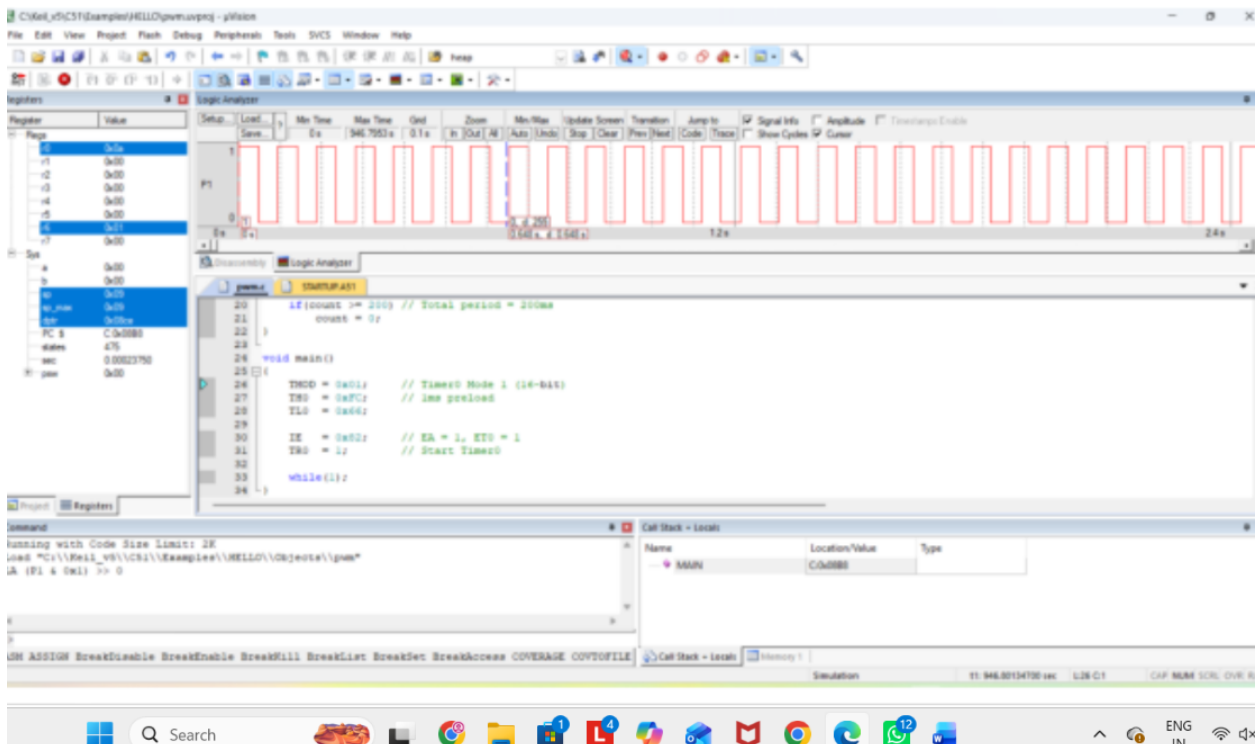
EXPERIMENT :16

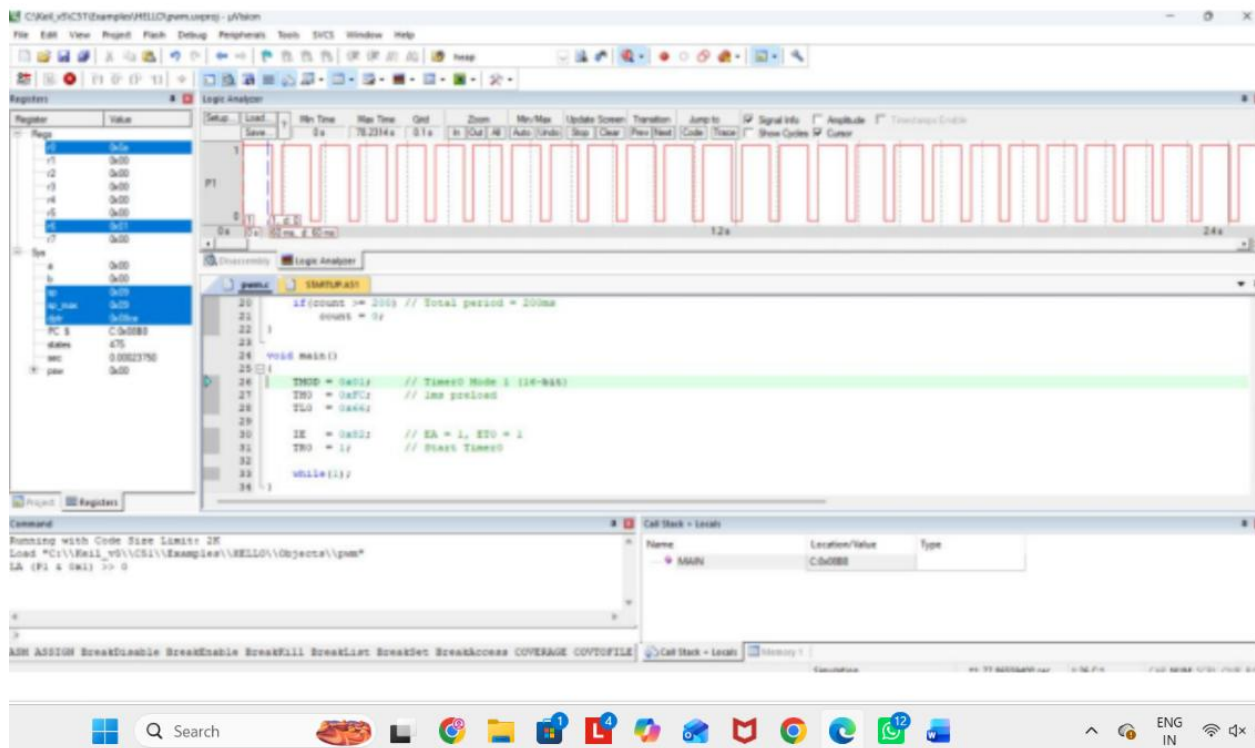


EXPERIMENT :17



EXPERIMENT :18





EXPERIMENT 24

```
jovi@DESKTOP-73175QB:~/systemc_test/build$ ./test_sys

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Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
jovi@DESKTOP-73175QB:~/systemc_test/build$ |
```

EXPERIMENT :25

```
jovi@DESKTOP-73175QB:~/systemc_test/build$ cd ~/systemc_test
jovi@DESKTOP-73175QB:~/systemc_test$ cd build
jovi@DESKTOP-73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/jovi/systemc_test/build
jovi@DESKTOP-73175QB:~/systemc_test/build$ make
[ 33%] Built target and_gate
[ 66%] Built target producer_consumer
[100%] Built target test_systemc
jovi@DESKTOP-73175QB:~/systemc_test/build$ ./producer_consumer

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Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Produced: 4
Consumed: 3
Produced: 5
Consumed: 4
Consumed: 5
jovi@DESKTOP-73175QB:~/systemc_test/build$ |
```

EXPERIMENT 26:

```
jovi@DESKTOP-73175QB:~/systemc_test/build$ cd ..
jovi@DESKTOP-73175QB:~/systemc_test$ ls
CMakeLists.txt and_gate.cpp build producer_consumer.cpp test_systemc.cpp test_systemcc.cpp
jovi@DESKTOP-73175QB:~/systemc_test$ nano tlm.cpp
jovi@DESKTOP-73175QB:~/systemc_test$ nano CMakeLists.txt
jovi@DESKTOP-73175QB:~/systemc_test$ cd build
jovi@DESKTOP-73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/jovi/systemc_test/build
jovi@DESKTOP-73175QB:~/systemc_test/build$ make
[ 25%] Built target and_gate
[ 50%] Built target producer_consumer
[ 75%] Built target test_systemc
[ 87%] Building CXX object CMakeFiles/tlm.dir/tlm.cpp.o
[100%] Linking CXX executable tlm
[100%] Built target tlm
jovi@DESKTOP-73175QB:~/systemc_test/build$ ./tlm

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Initiator sending: 555
Target received: 555
jovi@DESKTOP-73175QB:~/systemc_test/build$ |
```


EXPERIMENT:27

```
jovi@DESKTOP-73175QB:~/systemc_test$ nano CMakeLists.txt
jovi@DESKTOP-73175QB:~/systemc_test$ ls /opt/systemc
include lib share
jovi@DESKTOP-73175QB:~/systemc_test$ ls /opt/systemc/lib
cmake libsystemc.so libsystemc.so.2.3 libsystemc.so.2.3.4
jovi@DESKTOP-73175QB:~/systemc_test$ nano binary_semaphore.cpp
jovi@DESKTOP-73175QB:~/systemc_test$ nano CMakeLists.txt
jovi@DESKTOP-73175QB:~/systemc_test$ nano CMakeLists.txt
jovi@DESKTOP-73175QB:~/systemc_test$ cd build
jovi@DESKTOP-73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.1s)
-- Generating done (0.0s)
-- Build files have been written to: /home/jovi/systemc_test/build
jovi@DESKTOP-73175QB:~/systemc_test/build$ make
[ 20%] Built target and_gate
[ 40%] Built target producer_consumer
[ 60%] Built target test_systemc
[ 80%] Built target tlm
[ 90%] Building CXX object CMakeFiles/binary_semaphore.dir/binary_semaphore.cpp.o
[100%] Linking CXX executable binary_semaphore
[100%] Built target binary_semaphore
jovi@DESKTOP-73175QB:~/systemc_test/build$ ./binary_semaphore

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Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
jovi@DESKTOP-73175QB:~/systemc_test/build$
```

EXPERIMENT :28

```
[ 33%] Built target producer_consumer
[ 50%] Built target test_systemc
[ 66%] Built target tlm
[ 83%] Built target binary_semaphore
[ 91%] Building CXX object CMakeFiles/counting_semaphore.dir/counting_semaphore.cpp.o
[100%] Linking CXX executable counting_semaphore
[100%] Built target counting_semaphore
jovi@DESKTOP-73175QB:~/systemc_test/build$ ./counting_semaphore
-bash: ./counting_semaphore: No such file or directory
jovi@DESKTOP-73175QB:~/systemc_test/build$ ./counting_semaphore

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Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0
In file: /home/jovi/systemc-2.3.4/src/sysc/kernel/sc_object_manager.cpp:153
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: /home/jovi/systemc-2.3.4/src/sysc/kernel/sc_object_manager.cpp:153
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
jovi@DESKTOP-73175QB:~/systemc_test/build$
```

EXPERIMENT:29

```
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
jovi@DESKTOP-73175QB:~/systemc_test/build$ cd ..
jovi@DESKTOP-73175QB:~/systemc_test$ nano priority_interrupt.cpp
jovi@DESKTOP-73175QB:~/systemc_test$ nano CMakeLists.txt
jovi@DESKTOP-73175QB:~/systemc_test$ cd build
jovi@DESKTOP-73175QB:~/systemc_test/build$ cmake ..
-- Configuring done (0.0s)
-- Generating done (0.0s)
-- Build files have been written to: /home/jovi/systemc_test/build
jovi@DESKTOP-73175QB:~/systemc_test/build$ make
[14%] Built target and_gate
[28%] Built target producer_consumer
[42%] Built target test_systemc
[57%] Built target tlm
[71%] Built target binary_semaphore
[85%] Built target counting_semaphore
[92%] Building CXX object CMakeFiles/priority_interrupt.dir/priority_interrupt.cpp.o
[100%] Linking CXX executable priority_interrupt
[100%] Built target priority_interrupt
jovi@DESKTOP-73175QB:~/systemc_test/build$ ./priority_interrupt

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Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
jovi@DESKTOP-73175QB:~/systemc_test/build$
```



EXPERIMENT 30:

```
-- Build files have been written to: /home/jovi/systemc_test/build
jovi@DESKTOP-73175QB:~/systemc_test/build$ make
[ 12%] Built target and_gate
[ 25%] Built target producer_consumer
[ 37%] Built target test_systemc
[ 50%] Built target tlm
[ 62%] Built target binary_semaphore
[ 75%] Built target counting_semaphore
[ 87%] Built target priority_interrupt
[ 93%] Building CXX object CMakeFiles/rtos_scheduler.dir/rtos_scheduler.cpp.o
[100%] Linking CXX executable rtos_scheduler
[100%] Built target rtos_scheduler
jovi@DESKTOP-73175QB:~/systemc_test/build$ ./rtos_scheduler

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Task1 Running at 0 s
Task2 Running at 0 s
Task1 Running at 1 s
Task2 Running at 1 s
Task1 Running at 2 s
Task2 Running at 2 s
Task1 Running at 3 s
Task2 Running at 3 s
Task1 Running at 4 s
Task2 Running at 4 s
Task1 Running at 5 s
Task2 Running at 5 s
jovi@DESKTOP-73175QB:~/systemc_test/build$
```

